

The Region-Soundness Gap in GCS Planning: a First Measurement, Its Consequences, and a Certified Layer That Survives It

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Abstract

Every planner in the graphs-of-convex-sets (GCS) family rests on one sentence: *points inside the regions are collision-free*. The flagship paper acknowledges, in a single sentence, that its fast region generator “does not provide a rigorous certification, but ... appears to be very reliable in practice”; of the six follow-on planning papers we examined in full, five do not mention the assumption at all and the sixth confines plans to the region union it calls “certified” C-Free—in scare quotes, and grown by the same uncertified generator; no published work measures the consequences. We report the first measurement. The sentence holds for a curated demonstration library (0.3% contamination) and fails for scaled standard practice: 3.2% of interface samples in collision, 62% of solvable *task-query* reference optima physically colliding (up to 51% of a path’s length, penetrations to 91 mm), returned identically—and silently—by search- and relaxation-based planners alike. This note is deliberately not an accusation: every theorem in the family is correct relative to its input, and the affected papers’ own experimental domains largely dodge the issue. We supply the missing measurement, and a planning layer that survives it: sampled interface roadmaps with an informed filter, lazy physical validation with in-region repair, convex corridor polish, and a continuous clearance certificate—together with the small collection of lemmas that make the layer sound, including the observation that ε -optimality certificates *degrade gracefully* under contamination while unvalidated geometric costs simply become false.

1 The assumption and its paper trail

Let $\mathcal{F} \subset \mathbb{R}^d$ be the collision-free configuration space and $\mathcal{U} = \bigcup_{i=1}^n X_i$ a union of convex regions produced by an IRIS-family generator. Every GCS planner argues

$$q(t) \in \mathcal{U} \quad \forall t \quad \implies \quad q(t) \in \mathcal{F} \quad \forall t, \tag{1}$$

i.e. *containment implies safety*, which is exactly the claim $\mathcal{U} \subseteq \mathcal{F}$. How the literature treats this claim (verified against the full texts):

paper			treatment of $\mathcal{U} \subseteq \mathcal{F}$
GCS	flagship	(Sci. Robotics)	one sentence: the fast generator “does not provide a rigorous certification, but ... appears to be very reliable in practice”; no measurement; outputs not re-checked
GCS*	(WAFR’24)		not discussed; “sound” absent from the text; guarantees stated relative to the mathematical program
Multi-query SWP	(2025)		not discussed; IRIS-NP used as a black box (“we produce a convex decomposition of the collision-free configuration space”)
IxG*	(2024)		assumption only (sets “capture the free and safe planning space”)
GHOST	(AAAI’26)		not discussed
TANGO	(ICRA’26)		not discussed; one example “assumes no self-collision”
GGCS	(RSS’23/IJRR’25)		regions from the flagship’s uncertified NP grower (cites the C-IRIS paper but runs only its growth subroutine, 30 s/region); plans confined to “certified” C-Free—scare quotes theirs, meaning coverage, not validity; concludes trajectories are “collision-free at every point in time”; validity never measured
IrisZo	(generator)		honest: probabilistic contract—at confidence $1 - \delta$, colliding volume fraction $\leq \varepsilon$
C-IRIS	(generator)		exists precisely because certified regions were missing (SOS proofs; expensive)

The generation side states a *probabilistic* contract; the planning side consumes it as a *deterministic* one, and no published work quantifies the difference. Nobody is wrong; a measurement is missing.

2 Why pockets must exist

The pocket in Fig. 2 requires no one to make a mistake. It is the end product of a five-step chain in which every step is a mathematical fact.

Step 1: soundness is a universally quantified statement; collision checking is a pointwise oracle. The generator must deliver

$$P = \{x : Ax \leq b\} \subseteq \mathcal{F} \iff \forall q \in P : q \in \mathcal{F}, \quad (2)$$

a claim over a continuum. The only available primitive is a membership oracle for single configurations. Between the two sits a logical gap:

$$\underbrace{q_1 \in \mathcal{F} \wedge \dots \wedge q_M \in \mathcal{F}}_{\text{finite conjunction (checkable)}} \not\Rightarrow \underbrace{\forall q \in P : q \in \mathcal{F}}_{\text{universal claim (needs proof)}}. \quad (3)$$

Any colliding subset whose detection probability under the sampling measure is below $1/M$ is *logically consistent* with M clean samples. Finite sampling cannot exclude small pockets in principle.

Why pockets MUST exist: (a) convexity makes cuts complete proofs; (b) C-space non-convexity makes them local; (c) sampling cannot certify the universal claim, so the contract tolerates $\epsilon > 0$; (d) optimization provably exploits whatever remains.

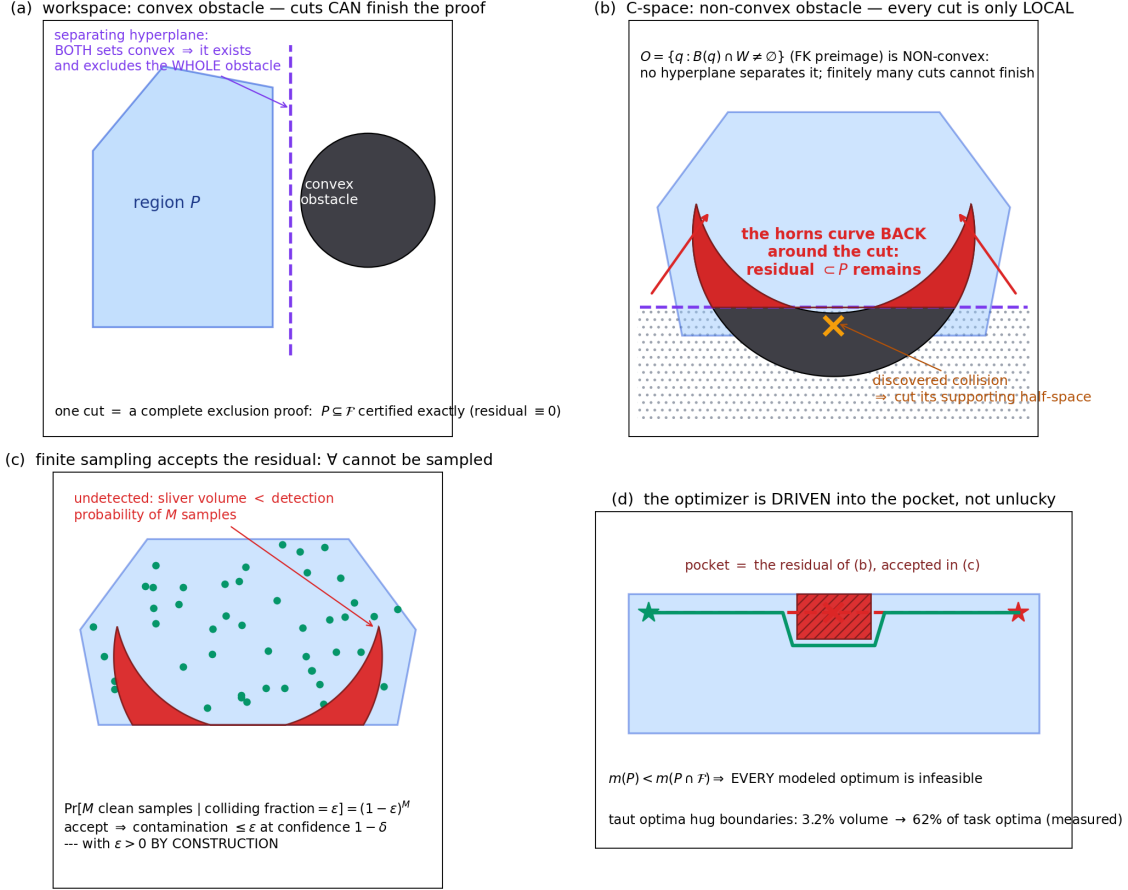


Figure 1: The derivation as a picture. (a) With convex obstacles the separating-hyperplane theorem makes every cut a complete exclusion proof: workspace IRIS is exactly sound. (b) A C-space obstacle is the preimage of a convex set under forward kinematics and is non-convex; a cut at the discovered collision excludes only a supporting half-space, and the horns curve back around the cut—the red residual inside the region is computed by actual polygon clipping, not drawn. (c) Finite sampling cannot certify the universal claim: interior samples miss the boundary slivers and the region is accepted under a contract that tolerates $\epsilon > 0$ by construction. (d) Whenever the through-pocket shortcut is shorter, every optimum of the modeled problem is infeasible (Prop. 1)—the measured 3.2% $\rightarrow$ 62% amplification is this effect.

Step 2: in configuration space, hyperplane cuts cannot close the gap. Why does workspace IRIS with convex obstacles *not* suffer from this? Because there the separating-hyperplane theorem applies: a convex region disjoint from a convex obstacle admits a hyperplane strictly separating the two, and one cut removes the *entire* obstacle from the region. Each cut is a complete exclusion proof (our 2-D testbed measures penetration exactly zero). In configuration space the hypothesis fails. For a convex workspace obstacle W and robot geometry $B(q)$ posed by forward kinematics, the C-obstacle is the preimage

$$O = \{q : B(q) \cap W \neq \emptyset\}, \quad (4)$$

and forward kinematics is built from products of rotations—trigonometric polynomials in the joint angles. Preimages of convex sets under such nonlinear maps are in general non-convex (the classical “banana” C-obstacle of a two-link arm). A hyperplane cannot separate a convex region from a non-convex obstacle; a cut placed at a *discovered* colliding configuration excludes only a local supporting half-space, and the parts of the obstacle that curve back around the cut remain inside the region. Excluding a non-convex set with finitely many cuts is impossible in general (a C-shaped obstacle requires cuts along its entire inner boundary). A genuine universal proof requires algebraic positivity certificates (Positivstellensatz / sums-of-squares in a rational parametrization)—which is precisely C-IRIS, whose existence is the field’s own acknowledgment that cuts cannot certify, and whose cost is why scaled pipelines do not use it.

Step 3: practice therefore degrades to statistical acceptance—with $\varepsilon > 0$ written into the contract. Unable to prove, the generator tests: accept the region after M consecutive clean samples. The mathematical content is the Bernoulli bound of Lemma 5,

$$\Pr[\text{accept} \mid \text{colliding fraction} > \varepsilon] \leq (1 - \varepsilon)^M \leq e^{-\varepsilon M} \leq \delta \quad \text{for } M \geq \frac{\ln(1/\delta)}{\varepsilon}, \quad (5)$$

and the contract *by construction* tolerates a colliding volume fraction up to $\varepsilon > 0$ (a zero- ε contract needs $M = \infty$). The shipped defaults are $\varepsilon = 1\%$, $\delta = 5\%$, and an alternation cap of 3 iterations—regions that hit the cap are admitted with the test unfinished. The pocket of Fig. 2 is not a breach of this contract; it is the contract.

Step 4: geometry stacks the residual exactly where planners work. Three effects push the tolerated residual into the same place. (i) Cuts are placed tangent at *discovered* collision points, so undiscovered slivers hug the polytope faces: the residual lives in the boundary shell. (ii) High-dimensional volume concentrates there: shrinking a d -dimensional convex body by a factor $1 - t$ leaves $(1 - t)^d$ of its volume, so a $t = 5\%$ shell in $d = 14$ holds $1 - 0.95^{14} \approx 51\%$ of the body—thin is not small—while interior-biased samplers under-sample precisely this shell. (iii) Interfaces $X_u \cap X_v$ are *double* boundary zones, the meeting of two obstacle-hugging shells—and interfaces are exactly where planners place waypoints. This is why our interface-biased measurement (3.2%) can exceed the volume contract (1%) without contradiction: it is a different, more operationally relevant measure of the same residual.

Step 5: the optimizer does not stumble into the pocket; it is driven there. Let $P_{\text{true}} = P \cap \mathcal{F}$ and let $m(S)$ denote the optimal path cost within S , so $P \supseteq P_{\text{true}}$ gives $m(P) \leq m(P_{\text{true}})$.

Proposition 1 (Optimizers exploit model surplus). *If $m(P) < m(P_{\text{true}})$, then every optimal solution of the modeled problem is infeasible for the true problem.*

Proof. An optimal modeled path costs $m(P) < m(P_{\text{true}})$, while every truly feasible path costs at least $m(P_{\text{true}})$. $\square$

Whenever the through-pocket shortcut is genuinely shorter, the exact solver of the modeled problem *must* return it—the more exact the solver, the more certainly. In Fig. 2 this is the gap $7.68 < 7.89$: its strict positivity is the proof that the geometric optimum uses the grown residual. A selection effect compounds the necessity: taut shortest paths hug region boundaries and cut corners—they patrol exactly the shell of Step 4. The measured amplification from 3.2% contaminated *volume* to 62% colliding task *optima* is this effect in the data, as is the factor-five gap between task queries (grasp poses riding the shell) and random queries (mostly interior).

The chain in one line.

$$\underbrace{\text{universal claims resist sampling}}_{(1)} + \underbrace{\text{non-convexity kills cuts}}_{(2)} \Rightarrow \underbrace{\varepsilon > 0 \text{ contract}}_{(3)} \xrightarrow{\text{geometry}} \underbrace{\text{residual in the shell}}_{(4)} \xrightarrow{\text{optimi}}$$

The pocket exists by contract (3), sits where geometry dictates (4), and is found because optimizers provably hunt for such surpluses (5). This is also why the remedy must be layered: runtime validation answers (5), a stricter enforced gate tightens (3), and only algebraic certification (C-IRIS) removes (2); step (1) is logic itself, and nothing gets around it.

3 The measurement

Protocol. Two 14/7-DOF libraries built with the standard pipeline (Drake IrisZo): a *curated* 7-DOF manipulator library (55 regions; task-covering seeds) and a *scaled* 14-DOF bimanual library (219 regions; rejection-sampled random seeds, generated with the generator’s *default* contract $\varepsilon = 1\%$, $\delta = 5\%$, `max_iterations`=3). Contamination is measured where planners place waypoints: samples drawn inside *interface polytopes* $X_u \cap X_v$, re-checked by the same collision checker that grew the regions (no model mismatch by construction). Planner outputs are audited by a four-step protocol: re-run the planner with the benchmark’s exact configuration and require bit-exact reproduction of the stored cost; re-solve the convex program whose optimum *is* the reported cost, keeping all waypoints (this argmin is the answer the planner commits to); validate that path by two independent mechanisms (the continuous certificate of Sec. 4.3 and 2,000 boolean probes); classify every violation by colliding geometry pair and penetration depth.

Findings.

	curated 7-DOF	scaled 14-DOF
interface-sample contamination	0.3%	3.2%
random-query reference optima colliding	—	2/16
task-query reference optima colliding	—	18/29 (62%)
worst penetration observed	—	91.3 mm
median worst-depth over colliding task pairs	—	39.4 mm
longest in-collision fraction of a path	—	51%

Three details matter. (i) On the flagship random query the stored reference cost is reproduced to every printed digit, our independent chain-program argmin returns that same cost, and the resulting path collides at 286 of 2,000 probe points with 63.5 mm worst penetration—eliminating “different path, same cost” explanations. The entire 16-query audit was independently re-run on a second machine: all verdicts and the exact colliding-probe counts (286 and 581) reproduce identically. One further query does not re-solve stably (run-to-run cost drift 1.7%, far above solver noise); the re-solved variant collides as well (gripper–gripper, -1.6 mm). We report it as *unstable-and-colliding* rather than counting it among the confirmed cases, which is why the table says 2/16. A full min-clearance sweep over all 18 colliding task-pair references (every cost reproduced to the last digit) shows these are not grazes: median worst penetration -39.4 mm, maximum -91.3 mm, and in every single case the violation is one contiguous mid-path interval while both endpoints are comfortably clear ($+38$ to $+100$ mm)—the named grasp configurations are valid; the *path* is what enters the shelves. (ii) Exposure depends on the *query distribution*: task queries (grasp poses hugging the shelves, whose corridors ride the grasp-seeded regions) collide five times more often than random

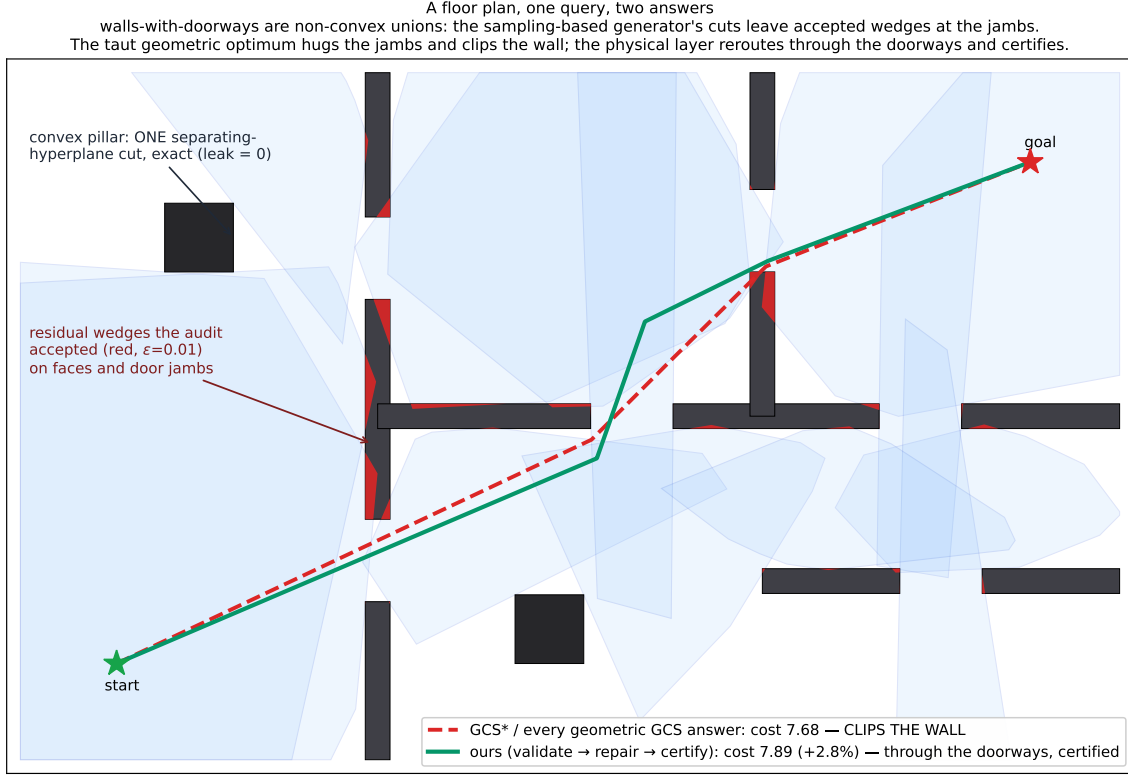


Figure 2: The whole story on one floor plan. Walls with doorways are *non-convex unions*; the same sampling-based generator (point oracle, supporting-half-plane cuts, Bernoulli ($\epsilon=1\%$, $\delta=5\%$) acceptance) that removes the convex pillars *exactly* (one projection-theorem cut each, measured leak 0) leaves 33 accepted residual wedges (red) on the wall faces and door jambs—computed by polygon clipping, not drawn. The taut geometric optimum (dashed red, cost 7.68)—the answer every GCS-family planner returns—hugs a jamb and clips the wall 4 cm below the door opening ($\times$). Our layer detects the violation, repairs, applies validated shortcutting, and certifies the green path: it passes through the actual doorway at +2.8% cost.

queries. The users’ actual workload is the worst case. (iii) Every mitigation short of regeneration fails on the scaled library: margin erosion leaves contamination flat ($3.4\% \rightarrow 2.6\%$ at $\delta = 0.05$ rad) while disconnecting queries; excision surgery spends 24,080 cuts, exhausts its budget on 105/219 regions, destroys 19% of portals and still leaves 1.25%—the contamination is *deep*, sitting inside overlap volumes, exactly as the generator’s truncated iterations would predict. On the curated library the same surgery converges completely ($0.3\% \rightarrow 0.05\%$, nothing lost). Contamination has two regimes, and three cheap diagnostics (interface sampling, erosion flatness, cut-budget exhaustion) identify the regime before effort is wasted.

Fairness. None of this invalidates any published theorem or experiment: GCS*’s guarantees are relative to its input sets and its own experiments use exact planar polyhedral decompositions; the flagship’s curated demonstrations sit in the regime where its “reliable in practice” sentence is measurably true. The gap opens where the field is heading: scaled, automatically generated C-space libraries.

4 The certified layer

Notation: query (s, g) , $s, g \in \mathcal{U}$; a path is a continuous curve in $\mathcal{U}$ with arc-length cost; c_{geom}^* is the optimal cost in $\mathcal{U}$ (the quantity all GCS planners compute); c_{phys}^* is the optimal cost in $\mathcal{U} \cap \mathcal{F}$ (the quantity a robot can execute). $\text{GCS}(\mathcal{A})$ denotes the shortest-path convex relaxation restricted to the subgraph induced by $\mathcal{A} \subseteq \{X_i\}$; its optimal value lower-bounds the best path cost within $\mathcal{A}$.

4.1 Admissible pruning and certificate migration

Lemma 1 (Admissible region pruning). *Let $\bar{c} \geq c_{\text{geom}}^*$ and $\ell(X) = \min_{x \in X} (\|x - s\| + \|x - g\|)$. If $\ell(X) > \bar{c}$, no geometrically optimal path visits X .*

Proof. A path through $x \in X$ has length $\geq \|x - s\| + \|x - g\| \geq \ell(X) > \bar{c} \geq c_{\text{geom}}^*$ by the triangle inequality. $\square$

Any lower bound $\hat{\ell}(X) \leq \ell(X)$ keeps the test admissible ($\hat{\ell}(X) > \bar{c} \Rightarrow \ell(X) > \bar{c}$); in high dimension we use the closed-form box bound $\hat{\ell}(X_u \cap X_v) = d(s, B_{uv}) + d(g, B_{uv})$ with $B_{uv} = \text{bbox}(X_u) \cap \text{bbox}(X_v) \supseteq X_u \cap X_v$, so the per-query filter costs a vector comparison, not a cone program.

Lemma 2 (Certificate migration). *Let $\mathcal{A} = \{X : \ell(X) \leq \bar{c}\} \cup \{X \ni s, g\}$ for a valid $\bar{c}$, and $\text{LB}_{\mathcal{A}}$ the optimum of $\text{GCS}(\mathcal{A})$. Then $\text{LB}_{\mathcal{A}} \leq c_{\text{geom}}^*$, so for any feasible cost $\hat{c}$ the subgraph gap $(\hat{c} - \text{LB}_{\mathcal{A}})/\hat{c}$ upper-bounds the true optimality gap. In practice we report $\max(\text{LB}_{\mathcal{A}}, \|g - s\|)$: both bounds are sound and on overlap-dense instances the relaxation bound can fall below the straight line.*

Proof. By Lemma 1, $\mathcal{A}$ contains every region of some optimal path, so the best cost within $\mathcal{A}$ equals c_{geom}^* , which the relaxation lower-bounds. $\square$

4.2 Contamination immunity

Lemma 3 (Graceful degradation). *Let UB_{phys} be the cost of any path validated to lie in $\mathcal{U} \cap \mathcal{F}$, and LB any sound lower bound for the geometric problem (Lemma 2). Then*

$$\text{LB} \leq c_{\text{geom}}^* \leq c_{\text{phys}}^* \leq \text{UB}_{\text{phys}},$$

so $\varepsilon = (\text{UB}_{\text{phys}} - \text{LB})/\text{UB}_{\text{phys}}$ is a valid certificate for the physical problem regardless of contamination.

Proof. $\mathcal{U} \cap \mathcal{F} \subseteq \mathcal{U}$ shrinks the feasible set, so $c_{\text{phys}}^* \geq c_{\text{geom}}^* \geq \text{LB}$; the validated path is feasible for the physical problem. $\square$

Remark 1. *Contamination therefore loosens our certificates (the chain stretches) but never falsifies them. The quantity that silently becomes false is the unvalidated geometric upper bound: the 62% of task-query reference costs above are exactly such UBs. This asymmetry—bounds that degrade versus numbers that lie—is the entire case for a physical layer.*

4.3 The continuous clearance certificate

Let $\phi(q) \geq 0$ denote the workspace clearance at configuration q : the minimum distance between any robot geometry and any obstacle (and between mutually-checked robot pairs), $\phi(q) = 0$ iff in contact.

Lemma 4 (Workspace Lipschitz bound). *Let r_i bound the distance from joint axis i to any point of the links it moves. For any robot point $p(q)$, $\|p(q') - p(q)\| \leq L \|q' - q\|$ with $L = (\sum_i r_i^2)^{1/2}$.*

Proof. $\|\partial p / \partial q_i\| \leq r_i$ for a revolute joint, so along the segment $q(t) = q + t(q' - q)$, $\|\dot{p}\| \leq \sum_i r_i |\dot{q}_i| \leq (\sum_i r_i^2)^{1/2} \|\dot{q}\|$ by Cauchy–Schwarz; integrate. $\square$

Proposition 2 (Certificate ball and conservative advancement). *If $\phi(q) > 0$ then every q' with $\|q' - q\| \leq \phi(q)/L$ is collision-free. Consequently, walking a segment with steps $\Delta_k \leq \phi(q_k)/L$ certifies it continuously: there is no resolution loophole. If the walk is driven below a floor $h_{\min}$ without positive clearance, the segment FAILS (colliding, or too thin to certify), with the offending interval reported.*

Proof. For any robot point and obstacle, $\text{dist}(p(q'), \mathcal{O}) \geq \phi(q) - \|p(q') - p(q)\| \geq \phi(q) - L\|q' - q\| > 0$ by Lemma 4; robot–robot pairs are identical with L replaced by the sum of the two arms’ constants. $\square$

For our platforms we use generously rounded reach-sum constants ($L = 4.0$ for the 7-DOF arm, 6.0 for the bimanual pair); conservative L only costs extra clearance queries (2.3 ms each), never soundness. A full path certifies in $\sim 10^2$ queries—versus the $\sim 10^6$ point checks a sampling planner spends per query on edge validation.

4.4 The planning pipeline

(i) Interface roadmap. Offline, one node per portal $P_{uv} = X_u \cap X_v$ at its Chebyshev center; edges join portals sharing a region, so every edge segment lies inside one convex region by construction. Online, A^* gives a corridor; batched interface sampling (equal budget per interface) refines it anytime.

Proposition 3 (Traversability rule). *A path may switch regions at x only where $\dim(X_u \cap X_v) \geq d - 1$. Interfaces of dimension $\leq d - 2$ (corners in 2-D, edges in 3-D) are pinch sets: regions meet there in measure zero and provide no passage. Samplers must therefore place boundary samples on the relative boundary of full-dimensional overlaps, and strictly inside degenerate contact faces.*

Violating this rule is not hypothetical: on a maze benchmark, a roadmap that admitted corner transitions returned cost 12.73 against a true optimum of 13.77—an “answer” 7.6% better than possible, caught precisely because it undercut a certified bound.

(ii) Informed filter. With incumbent c_{best} , interfaces with $\hat{\ell}(P) > c_{\text{best}}$ and nodes with $\|x - s\| + \|x - g\| > c_{\text{best}}$ are frozen (Lemma 1). Every node on the incumbent satisfies the bound by the triangle inequality, so *the incumbent is never lost and the cost curve is monotone by construction.*

(iii) Lazy validation and in-region repair. Only incumbent candidates are validated (Prop. 2). On failure at segment $[a, b]$ with witness region $X_r \ni a, b$: sample a via point $x \in X_r$; by convexity both replacement legs $[a', x], [x, b']$ (a', b' on the original segment) remain in X_r , so repair preserves containment; the repaired path is re-validated in full. What cannot be repaired is rejected and its edges blacklisted—the planner then either routes around the pocket or honestly reports that no certified path was found. On the contaminated library this honesty is the difference between our “no certified path” on 13/18 task pairs and the reference planners’ colliding “optima” on the same pairs.

(iv) Corridor polish. Given the corridor’s portal sequence $(P_{e_1}, \dots, P_{e_K})$ (extracted by a region-transition walk, so consecutive portals share a transit region and the chain below is feasible

by convexity), the restriction program

$$\min_{x_k \in P_{e_k}} \|x_1 - s\| + \sum_{k=1}^{K-1} \|x_{k+1} - x_k\| + \|g - x_K\| \quad (6)$$

places the waypoints exactly (a small SOCP, 10–60 ms). The polished path moves waypoints, so it re-enters the physical layer; if it cannot be validated or repaired, the pre-polish physical path stands. Division of labor, measured: raw sampled polylines carry +5.7% (7-DOF) to +41.5% (14-DOF) slack; polish collapses it to +1.3% / +9.6%. Sampling discovers corridors; convexity places waypoints; neither substitutes for the other.

(v) Output. UB = cost of the certified physical path; LB = $\max(\text{LB}_{\mathcal{A}}, \|g - s\|)$; ε per Lemma 3.

4.5 Fixing the library itself

Runtime repair treats symptoms. The measured cure hierarchy: (a) *excision surgery* (sample–bisect–cut until M consecutive clean samples) fully heals shallow contamination and maims deep contamination; (b) *margin erosion* is refuted in both regimes; (c) *regeneration* with a strict contract is the only cure for the deep regime. The acceptance test used in both (a) and (c) is the Bernoulli gate:

Lemma 5 (Audit gate). *If a region’s colliding volume fraction exceeds ε , the probability that M independent uniform samples are all clean is at most $(1 - \varepsilon)^M \leq e^{-\varepsilon M}$. Hence $M = \lceil \ln(1/\delta)/\varepsilon \rceil$ consecutive clean samples certify, with confidence $1 - \delta$, contamination $\leq \varepsilon$.*

This is the generator’s own statistical test, re-run to a stricter target ($\varepsilon = 0.1\%$, $\delta = 1\%$: $M = 4,605$) and—crucially—enforced as a *gate*: a region that fails is regrown or dropped, never silently admitted. The original scaled library was generated with $\varepsilon = 1\%$, $\delta = 5\%$ and an iteration cap of 3, with cap-truncated regions admitted without any gate; its measured 3.2% interface contamination is not an accident but an un-enforced contract.

5 What the layer delivers on the measured problem

claim	measurement
first solutions are fast	0.13–0.4 s median to first certified incumbent; 17–21 $\times$ earlier than the search baseline’s best (7-DOF library)
polish closes the sampling slack	+5.7% $\rightarrow$ +1.3% (7-DOF); +41.5% $\rightarrow$ +9.6% (14-DOF)
the physical layer is cheap	$\sim 10^2$ clearance queries (2.3 ms each) per incumbent; certificates in 0.02–0.06 s
it converts silent violations into decisions	head-to-head on all 29 task pairs: the reference delivers 21 answers failing continuous certification (18 probe-confirmed); ours delivers 15 certified + 14 honest no-paths + <i>zero</i> violations
correctness costs no speed	first certified incumbent median 1.14 s vs. 1.59 s for the reference’s unvalidated answer; ~ 200 clearance calls per certificate
correctness costs no optimality where the reference is right	on the 7 clean-reference pairs both arms solve, cost ratio 1.0000 (median and max): the certified answer reproduces the reference optimum digit-for-digit; the price of truth appears only where the reference is wrong (median +14%, range +4.8 to +56% on 8 rescued pairs)
certificates stay sound under contamination certification cost is governable	Lemma 3; measured gaps only widen a 60 s budget keeps 75% full certificates and cuts the p90 end-to-end time from 1,732 s to 71 s; survivor counts predict certification cost (log–log $r = 0.96$ –0.98) before it is paid

The registered prediction, resolved against us—and for the thesis. We pre-registered: on a strictly regenerated library (Lemma 5 enforced at generation, $\varepsilon = 0.1\%$, $\delta = 1\%$, $4\times$ particles, 10 iterations), most of the 13 unsolvable task pairs should become solvable. The run is complete (222 seeds, ~ 40 node-hours) and the prediction is refuted in the strongest possible direction: the gate accepts 47/222 seeds (21%; grasp seeds 3/22), half of the survivors only after regrowth retries, and the 47 accepted regions are *pairwise disjoint*—0 portals, verified independently by 1,081 pairwise LPs—versus 219 regions and 2,757 portals in the default library. The strict library supports no planning at all. Together with the C-IRIS composition experiment (a 2-DoF certified library fragments into 7 components and answers 0/30 queries even at $2.7\times$ the seed budget) and the erosion result above, three independent routes to volume-level certainty—a stricter sampling gate, SOS certificates, and uniform margins—all destroy the connectivity that planning needs before they deliver soundness where tasks live. Volume-level certainty and a usable region graph trade off

brutally; the path-level runtime certificate is the one mechanism in our study that keeps the graph and still returns only verified answers.

6 Summary

The GCS family’s guarantees are conditional on a sentence nobody had measured. Measured, the sentence splits: true for curated demonstrations, false at scale, and worst exactly on the task distribution users care about. The response need not be abandonment—it is a layer: validate what you return, repair what you can, refuse what you cannot, certify against bounds that survive the contamination, and regenerate libraries under an enforced statistical gate. Every claim above is a number in a benchmark we release, and none of it required a single competitor to be wrong.

Sources for Sec. 1: Marcucci et al., Science Robotics 2023 (§7.4); Chia et al., WAFR 2024; Morozov et al., arXiv:2507.10878; arXiv:2410.08909; Tang & Ma, AAAI 2026; arXiv:2603.11658; Werner et al. (IrisZo); Dai et al., IJRR 2024 (C-IRIS). Full quotations and per-paper verification notes in the project log.